

Red Sand

AP Chemistry · the back half

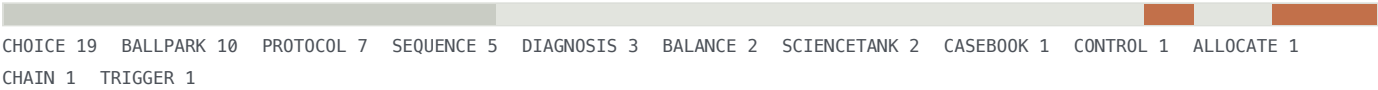
A propellant plant on Mars: modules buried in regolith along one track, under a butterscotch sky.

53	19	5	8/9	2	25 · 27 · 1
STOPS	CHOICE (36%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

One equation gap and it is the important one: **Q compared with K** is mentioned at eight stops and computed at none. That is the whole of the equilibrium half of AP Chemistry delivered as recognition. Nineteen CHOICE stops out of 45, and the three select-only concepts are all physical chemistry a student should be able to produce.

NOW The mix as it ships



CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
Q compared with K	reading which way a mixture will move from what is in it now	1 · PROTOCOL 2 · CHOICE 8 · CHOICE 10 · CHOICE 16 · CHOICE	real gap

0 of 1 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 1 is unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM	CONCEPT	STOPS
13	Sintering: losing surface to heat		30
22	The temperature at which a spontaneous reaction stops being one		10, 37, 50

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects.

FUN What this subject and this place can carry

SWEEP

STRONG

stop 8 or 21 · EQUIL

Temperature against conversion, and the curve turns over: hotter is faster and worse. "Why the hotter pass gave less" is currently a CHOICE, and it is the single best SWEEP available in the catalogue — the trade-off is the shape.

SERVES Q against K, Le Chatelier, and concept 22 — where a spontaneous reaction stops being one

NEEDS a broad plateau that beats a tall spike

HOLD STRONG

Hold reactor temperature inside a closing band while the feed composition drifts. On Mars the disturbance is free and authentic — a dust storm cuts the array and the band closes with it.

SERVES kinetics against equilibrium, felt rather than compared

NEEDS disturbances integrated with the control untouched, so a do-nothing run fails

TRIAL STRONG

the track

The place is nine modules buried in regolith **along one track**, with an ascent vehicle on a pad and a gauge that fills as the campaign does. A driven start-up order — power, then electrolysis, then the reactor — is the plant’s real dependency graph.

SERVES process order and the plant as one system

RISK world work; and the vehicle bay already exists, which is half the cost gone

PROBE STRONG

a new PHASE or ELEC stop

Take readings module by module along the track and find where conversion falls off. The linear site makes the chain physical instead of diagrammatic.

SERVES concept 27, vapour pressure, and reading a cryogenic tank by it

NEEDS one honest break that is not the worst number

SPOT MEDIUM

GIBBS · Plant Control

A load-shed instruction replaced mid-sol. Genuine — but the game already carries an ALLOCATE on exactly the dust-storm power budget, so this risks teaching the same thing twice.

SERVES operating discipline

RISK overlaps the sol-12 ALLOCATE

BELT DEFER

A propellant plant has no sorting line, and inventing one to host a format is how a place stops being a place. Defer unless a real sort turns up in the process.

LOB REJECT

The ascent vehicle is the obvious hook and it is the wrong one: a launch is computed here, in a course about computing things.

SLATES **Three appetites — the campaign is zero-sum**

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs redsand --snapshot before.json , then --against before.json .

Minimum honest 3 STOPS · ONE SITTING				
8	d3	EQUIL	CHOICE → SWEEP	Why the hotter pass gave less the hotter pass, as a curve you build
2	d1	KINET	CHOICE → CLOUD	Rate is not yield rate against yield, as two distributions that do not move together
36	d10	PHASE	CHOICE → HOLD	The tank pressure that is not a leak tank pressure held while the cold end warms
CHOICE 19 → 16 (30%) · moves 25·27·1 → 22·28·3 · effective formats 5.0 → 6.2 · 2 new operate stops, 1 of them arcade				

Balanced 6 STOPS · THE RECOMMENDATION

8	d3	EQUIL	CHOICE → SWEEP	Why the hotter pass gave less
2	d1	KINET	CHOICE → CLOUD	Rate is not yield
36	d10	PHASE	CHOICE → HOLD	The tank pressure that is not a leak
23	d7	GIBBS	CHOICE → TRIGGER	The reaction that likes being hot the temperature at which it stops being spontaneous, written as a rule
30	d9	KINET	CHOICE → PROBE	One line off the spent charge sintering found by reading the line module by module
39	d11	PHASE	CHOICE → BALANCE	The space left at the top the space at the top of the tank, closed as a ledger

CHOICE 19 → 13 (25%) · moves 25·27·1 → 19·30·4 · effective formats 5.0 → 7.5 · 3 new operate stops, 1 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

8	d3	EQUIL	CHOICE → SWEEP	Why the hotter pass gave less
2	d1	KINET	CHOICE → CLOUD	Rate is not yield
36	d10	PHASE	CHOICE → HOLD	The tank pressure that is not a leak
23	d7	GIBBS	CHOICE → TRIGGER	The reaction that likes being hot
30	d9	KINET	CHOICE → PROBE	One line off the spent charge
39	d11	PHASE	CHOICE → BALANCE	The space left at the top
1	d1	EQUIL	PROTOCOL → TRIAL	What the loop settles at the start-up order, driven along the track
16	d5	KINET	CHOICE → CONTROL	What a catalyst cannot do what a catalyst cannot do — change one thing and put it back
50	d14	ELEC	CHOICE → SPOT	What limits the current the current limit, under an instruction that changes

CHOICE 19 → 11 (21%) · moves 25·27·1 → 17·29·7 · effective formats 5.0 → 8.8 · 6 new operate stops, 3 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — tools/import-book.mjs books/redsand.yml redsand --verify. **4.** npm run traps: every instrument's trap must fire. **5.** npm run check redsand and npm run drive redsand, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from themes/redsand/content/ via engine/dev/curriculumDelivery.mjs; concept and equation lists authored in tools/syllabus.js. Format verdicts and slates are editorial. Run it with THEME=redsand npm run dev.